



AI-Based Customer Segmentation and Targeted Marketing

From Blanket Promotions to Behavioural Micro-Segments

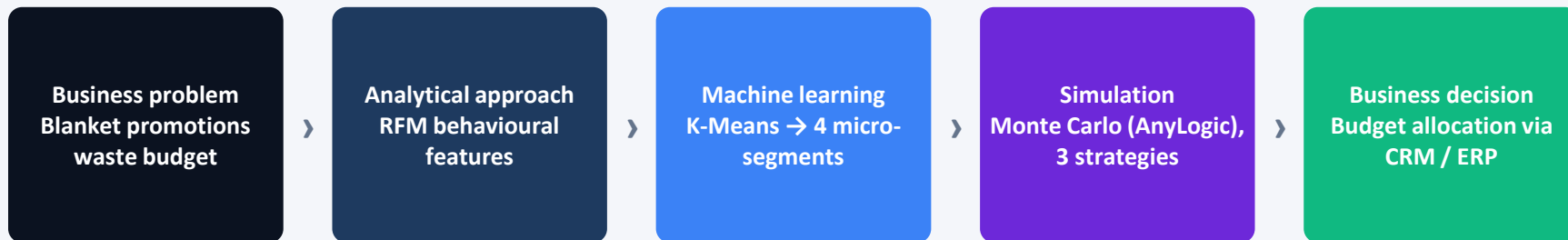
PREPARED BY:

Ehab Shoukry (ID: 24128379)

UNDER SUPERVISION OF:

Prof. Ahmed Elseddawy

One storyline: from raw data to a budget decision



Machine learning identifies WHO the customers are behaviourally; simulation shows HOW limited budget should be allocated under uncertainty.

Blanket promotions cannot keep up with e-commerce

Market saturation and rising customer acquisition costs (CAC) expose the limits of static segmentation and mass campaigns.

<1.5%

Low conversion

Untargeted mass marketing yields depressed click-through rates.

~15%

Revenue loss

Retailers lose approximately 15% of annual revenue to misaligned promotional campaigns.

Silos

Data locked in ERP

Rich transaction history sits in ERP systems without a predictive ML linkage.

CORE RESEARCH QUESTION

How can businesses integrate unsupervised K-Means clustering with supervised classification models to move from static demographic groupings to dynamic behavioural micro-segments?

Objective: static segments → behavioural intelligence

Dimension	Traditional approach	AI-enhanced approach (this framework)
Segmentation basis	Static demographics (age, region, gender)	Dynamic behavioural vectors (Recency, Frequency, Monetary)
Accuracy & granularity	Low; broad strokes, high misclassification	High; mathematically validated clusters (AUC 0.93–0.99*)
Processing speed	Manual / batch (quarterly / annual)	Automated, near real-time Python pipeline
Campaign conversion	Click-through rates <1.5%	Targeted conversion >4.5%** via personalised triggers

** AUC range is reported from the literature benchmark (Xiahou & Harada, 2022). ** Target outcome, not an observed result.*

K-Means model

Groups customers by behaviour (RFM) into micro-segments.

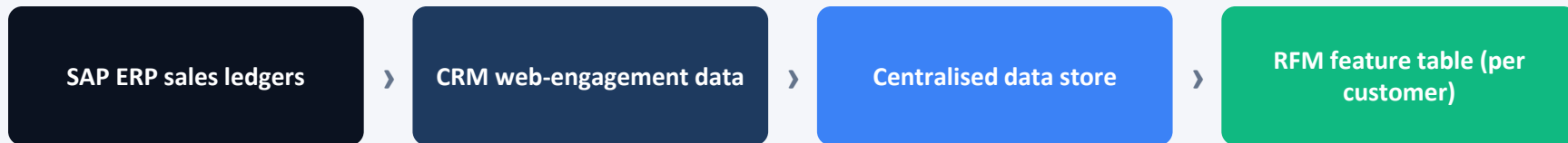
Monte Carlo simulation

Tests promotion-budget strategies under uncertainty.

Decision framework

Links segments, simulation and CRM / ERP to support management budget decisions.

Data & context: ERP transactions meet CRM engagement

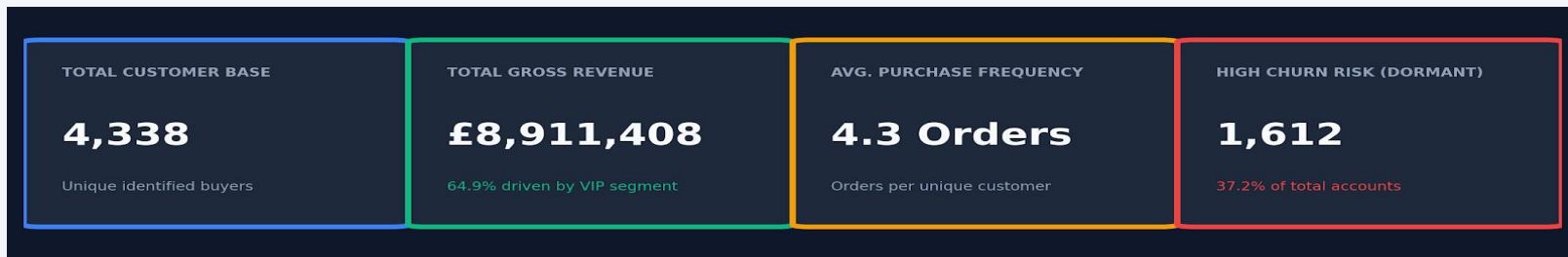


Transaction fields

InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country (online-retail transaction data)

Customer behaviour variables

Recency, Frequency, Monetary (TotalSpend); supporting customer file also holds Age, AnnualIncome and Segment (1,000-row sample)



Analysed base: 4,338 unique identified buyers, £8,911,408 gross revenue

RFM turns raw transactions into behavioural features



Recency

How recently a customer last purchased



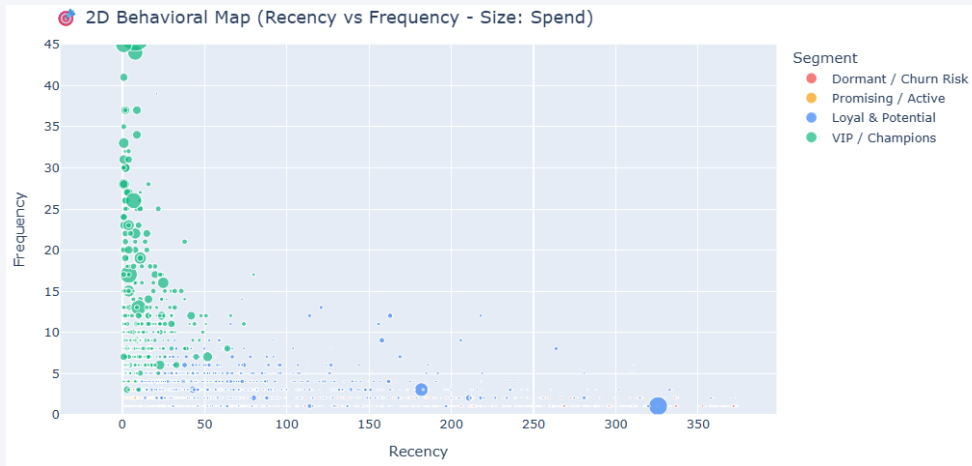
Frequency

How often the customer orders (avg. 4.3 orders per customer)



Monetary

How much the customer spends in total



Behavioural map: Recency vs Frequency (bubble size = spend). Active, high-frequency buyers cluster at low Recency.

Data-preparation detail (cleaning rules, outlier handling): [DATA REQUIRED]

Why unsupervised learning? Why K-Means?



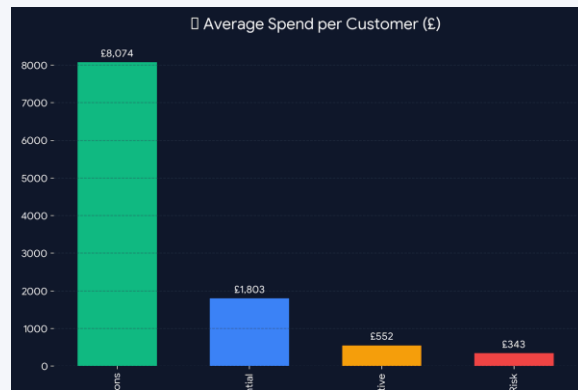
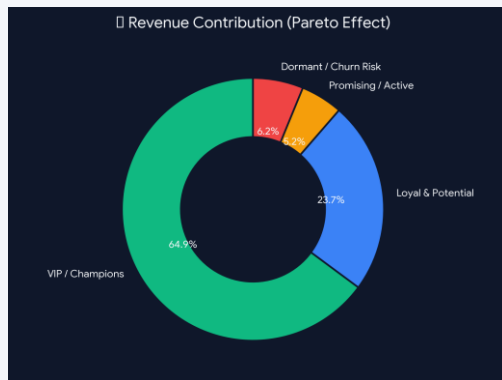
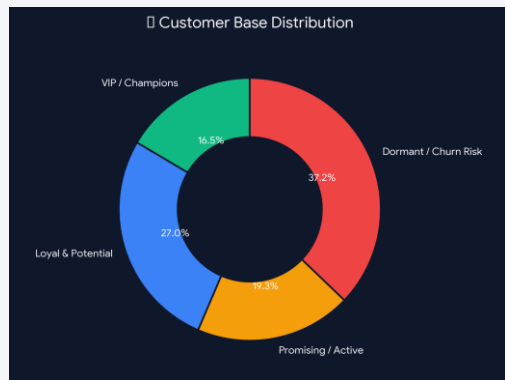
What was decided

Optimal K determined with Elbow Method and Silhouette analysis; result: 4 micro-segments (VIP / Champions, Loyal & Potential, Promising / Active, Dormant / Churn Risk). Project-specific elbow / silhouette values: [DATA REQUIRED]

Literature benchmarks (not this project's results)

Elbow: WCSS slopes at K=4 (Putri et al., 2026). Silhouette peaks at 0.10 for K=2, below 0.07 for higher K. Post-segmentation SVM AUC: 0.992 / 0.966 / 0.927 (Xiahou & Harada, 2022).

Four micro-segments: the Pareto effect in the customer base



Segment	Customers	Revenue	Revenue share	Avg spend
VIP / Champions	16.5%	£5.78M	64.9%	£8,074
Loyal & Potential	27.0%	£2.11M	23.7%	£1,803
Promising / Active	19.3%	£461K	5.2%	£552
Dormant / Churn Risk	37.2% (1,612)	£553K	6.2%	£343

Bars left→right: VIP, Loyal, Promising, Dormant. Recency / frequency values per segment: [DATA REQUIRED]

Executive dashboard: analytics management can monitor

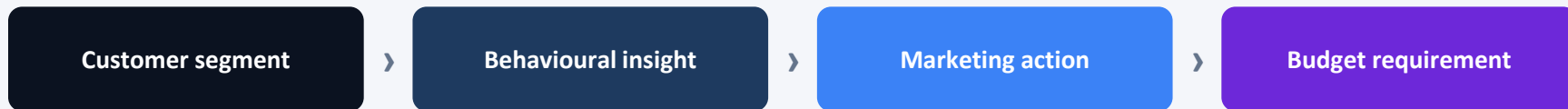


KPI strip: Customer base, revenue, frequency and churn exposure at a glance

Segment mix & revenue: Shows how few customers generate most revenue

Behavioural map: Recency vs Frequency view of every customer group

Segmentation alone is not a decision: tie it to budget



Segment	Share	Revenue	Recommended AI / CRM action	Budget
VIP / Champions	16.5%	£5.78M	Dedicated concierge, early product drops, retention rewards	→ simulation
Loyal & Potential	27.0%	£2.11M	Cross-selling; upsell premium tiers (target: +18% LTV lift)	→ simulation
Promising / Active	19.3%	£461K	Onboarding campaigns; next-order discount triggers within 14 days	→ simulation
Dormant / Churn Risk	37.2%	£553K	Automated win-back email flows; high-discount clearance incentives	→ simulation

Targeted outcome (dashboard, not yet observed): reallocating budgets by this matrix should cut churn by 15% and lift conversion from 1.5% to >4.5%. Budget amounts: [DATA REQUIRED]

CRM / ERP integration: operationalising the model



↻ Feedback loop: results return to the data store; cluster centroids are retrained quarterly.

Enablers and risks (from Task 3)

Requires clean, centralised ERP↔CRM pipelines (data governance) and automated API connectors between Python scoring and CRM tools.

Why simulation? Limited budget, uncertain response

Given limited promotional resources, how should the budget be allocated across customer segments?

Segments tell us who, not how much

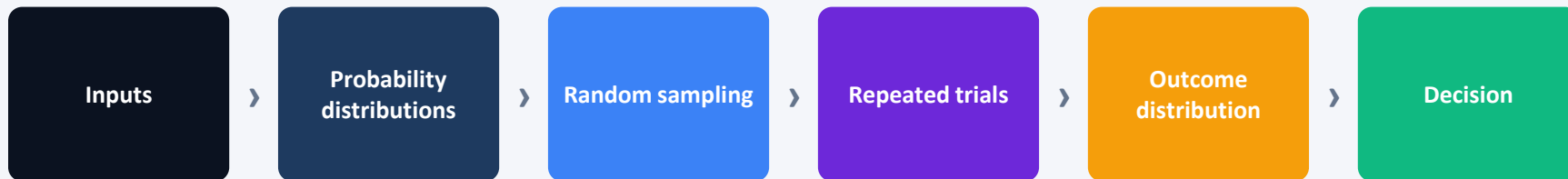
K-Means shows which customers behave alike. It does not say what a campaign will return once redemption, response and costs are uncertain, or which strategy is worth funding.

Three strategies compared

- S0 Mass promotion
- S1 Segment playbook
- S2 Value-prioritised

Each is simulated repeatedly, so management sees a distribution of outcomes rather than a single guess.

Monte Carlo methodology: decisions under uncertainty



Model inputs (AnyLogic)

redemption, response, targeting, contactCost, platformCost, perkCost, margin

Simulation set-up

Customers [1000] agents; 10,000 samples per strategy (dS0, dS1, dS2); net outcome per strategy (netS0–S2).

Not documented in source

Distribution types and parameters; units of the net outcome; budget per scenario: [DATA REQUIRED]

Simulation process: the AnyLogic model (Sim v1)



Model canvas before the run.

Parameters

redemption, response, targeting, costs, margin

Strategy outputs

dS0 / dS1 / dS2 histograms, netS0–S2

Colour code

S0 Mass promotion (orange), S1 Segment playbook (yellow), S2 Value-prioritised (purple)

Scenarios tested: three promotion strategies



Mass promotion

Blanket approach: the same promotion for all customers (baseline).

Budget / allocation / assumptions: [DATA REQUIRED]



Segment playbook

Segment-specific actions following the K-Means decision matrix.

Budget / allocation / assumptions: [DATA REQUIRED]



Value-prioritised

Budget prioritised toward higher-value segments.

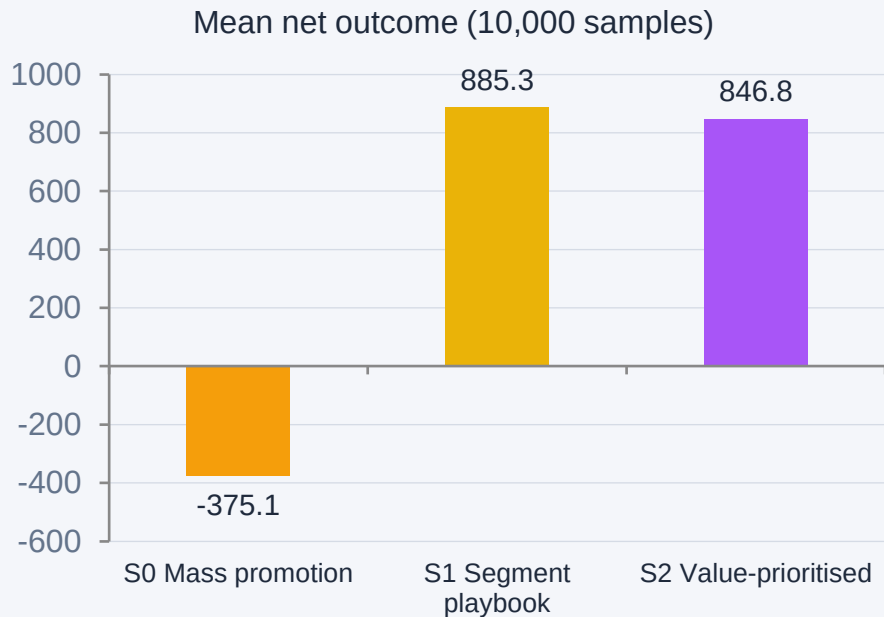
Budget / allocation / assumptions: [DATA REQUIRED]

Model readout at run completion (as displayed; units and definitions to be confirmed)

redemption	response	targeting	contactCost	platformCost	perkCost	margin
0.351	0.838	1.043	0.303	410.77	1.464	0.379

Scenario descriptions are inferred from the strategy names shown in the model; exact rules are not in the source decks.

Results: targeted strategies beat mass promotion



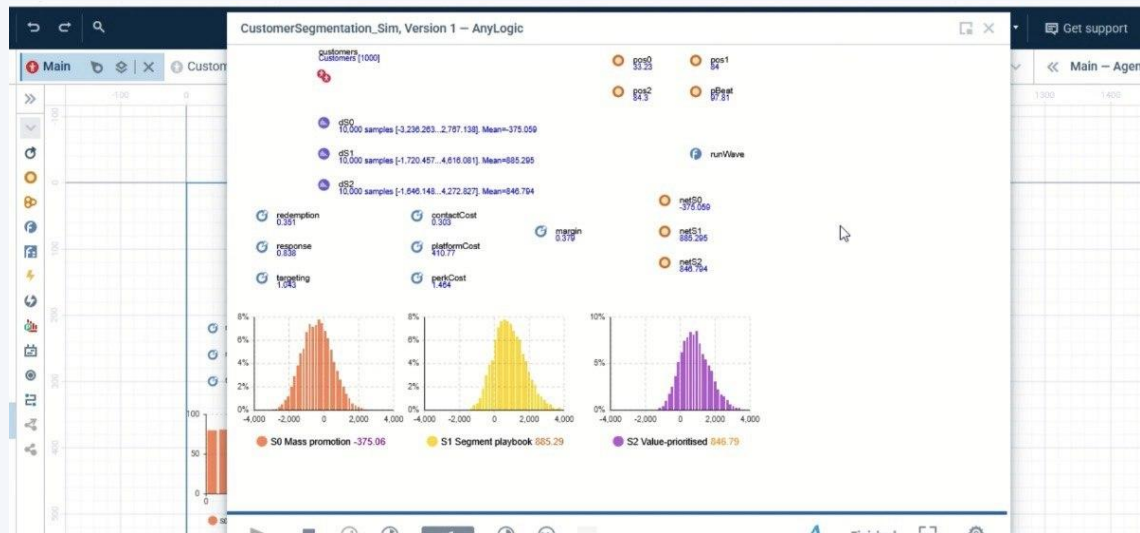
Strategy	Range (10,000 samples)	Mean
S0 Mass	-3,236.263 ... 2,767.138	-375.059
S1 Playbook	-1,720.457 ... 4,616.081	885.295
S2 Value-prio.	-1,646.148 ... 4,272.827	846.794

What it tells management

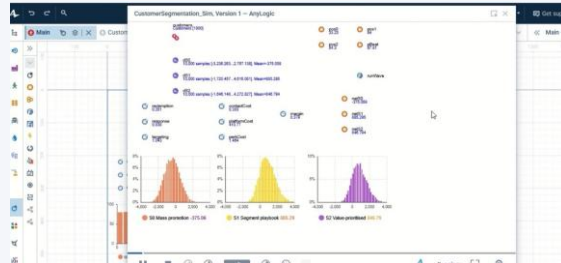
Mass promotion averages a loss; both segment-based strategies average a gain. S1 has the higher mean, S2 a narrower range of outcomes.

Units of net outcome: [DATA REQUIRED]

Simulation output: distributions of outcomes



Finished run: S0 -375.06 | S1 885.29 | S2 846.79



Run in progress

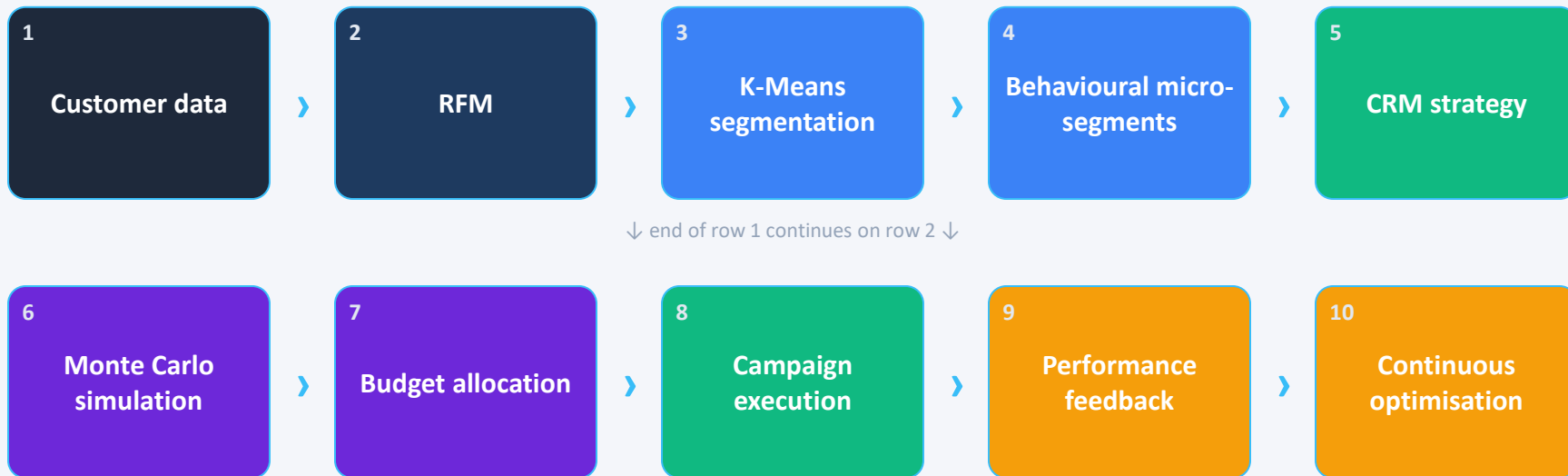
Reading the histograms

S0 is centred below zero; S1 and S2 are centred above zero, showing outcome uncertainty for each strategy.

Other model outputs shown (pos0 33.23, pos1 84, pos2 84.3, pBeat 97.81)

Likely probability-type readouts; definitions to be confirmed by the author before presenting.

Integrated decision framework



Raw data → intelligence → segmentation → simulation → decision → feedback

AI impact: before vs after adopting the framework

Dimension	Before AI (blanket approach)	After AI (this framework)
Campaign targeting	One-size-fits-all blanket promotion	4 behavioural micro-segments (K-Means)
Campaign conversion	<1.5% click-through	>4.5% targeted conversion (dashboard target)
Analysis turnaround	~40 hours of manual spreadsheet work	Under 5 minutes via Python pipeline
Revenue at risk	~15% of annual revenue lost to misaligned promotions	Targeted 15% reduction in churn
Segment visibility	Static demographics, quarterly/annual updates	Live dashboard, near real-time RFM scoring

Reading the comparison

Conversion, churn-reduction and LTV figures are the framework's stated targets, not yet observed outcomes; turnaround time and segment count are reported/observed results.

Business value: observed results vs expected benefits

OBSERVED (data / model output)

- 16.5% of customers generate 64.9% of revenue (£5.78M)
- 37.2% (1,612 accounts) are dormant but hold only 6.2% of revenue (£553K)
- Simulation: mean net outcome –375.06 (mass) vs 885.29 (playbook) and 846.79 (value-prioritised)
- Reported: manual analysis cut from 40 hours to under 5 minutes

EXPECTED / TARGETED (not yet validated)

- Conversion from <1.5% to >4.5%
- Churn reduced by 15%
- +18% LTV lift in Loyal & Potential
- Less reliance on blanket discounting
- Scalable CRM / ERP framework

Task 3 states conversion "tripled"; the dashboard presents it as a target, so it is shown here as targeted.

Recommendations

Short term

- Replace blanket discounts for VIPs with concierge retention
- Trigger automated win-back flows for the 37.2% dormant segment
- Apply the decision matrix by segment

Medium term

- Build automated API connectors between Python scoring and CRM
- Establish clean ERP↔CRM data governance
- Retrain cluster centroids quarterly

Long term

- Validate simulation with real campaign results [DATA REQUIRED]
- Add supervised churn models (Random Forest / SVM)
- Run a continuous feedback loop for budget re-allocation

Limitations & future development

Limitations

- Elbow, silhouette and AUC figures come from literature, not this model
- Simulation distributions, units and scenario budgets are undocumented
- Results are simulated; real-world validation is needed
- Behavioural variables limited to RFM so far
- Files provided are 1,000-row samples
- ERP↔CRM integration and governance effort

Future development

- Supervised churn prediction on cluster labels
- Clickstream behaviour added to ERP data
- Real-time campaign triggers pushed to CRM
- Live A/B testing to calibrate simulation inputs
- Scheduled model retraining

Final conclusion

K-Means identifies who customers are behaviourally

Monte Carlo simulation shows how resources can be allocated under uncertainty

CRM & ERP workflows provide the path to operationalise the insight

Machine learning finds the segments; simulation turns them into a budget decision.



Thank You

PREPARED BY: Ehab Shoukry (ID: 24128379)

SUPERVISED BY: Professor Ahmed El-Sedawwy